STEM SMART Physics 21 - Friction >

Thinking About Friction

Subject & topics: Physics | Mechanics | Statics **Stage & difficulty:** A Level P2

*This problem involves **friction** , which is not covered in some Physics A Levels. For more information please check with your teacher.*

Two people are trying to push a heavy box of mass $750\,\mathrm{kg}$ on a rough, horizontal surface.

Part A

Calculating a frictional force

The first person tries to push the box, applying a horizontal force of $150\,\mathrm{N}$ but they can't move it. Draw a free body diagram of the forces acting on the box, and state the magnitude of the frictional force acting on it. Give your answer to 2 significant figures.

Part B

Nature of a frictional force

Now the second person also applies a force of $150\,\mathrm{N}$ to the box, but it still doesn't move. What does this tell us about the nature of the frictional force?

Part C

Number needed to move block

The maximum frictional force that can be applied to the block by the rough surface is $650\,\mathrm{N}$. How many people, each applying $150\,\mathrm{N}$ to the block, will it take to move the block?

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Friction 4

Step Up to GCSE Physics 32.4

Subject & topics: Physics | Mechanics | Dynamics **Stage & difficulty:** Year 9 C2, GCSE C1

A $90\,\mathrm{kg}$ mass was pushed hard and let go. It is now slowing down on a $\mu = 0.090$ flat surface. Calculate its deceleration. Use $g = 10\,\mathrm{N/kg}$ in this question.

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Friction 6

Step Up to GCSE Physics 32.6

Subject & topics: Physics | Mechanics | Dynamics **Stage & difficulty:** Year 9 C2, GCSE C2

Use $g = 10\,\mathrm{N/kg}$ in this question.

$4.0\,\mathrm{kg}$ on a flat surface has static $\mu = 0.40$ and dynamic $\mu = 0.20$.

Part A

Resting

Is $8.0\,\mathrm{N}$ enough to start the mass moving from rest?

- No
- Yes

Part B

Steady speed

Is $8.0\,\mathrm{N}$ enough to keep the mass moving at a steady speed?

- No
- Yes

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Friction 7

Step Up to GCSE Physics 32.7

Subject & topics: Physics | Mechanics | Dynamics **Stage & difficulty:** Year 9 C2, GCSE C2

Use $g = 10\,\mathrm{N/kg}$ in this question.

An $800\,\mathrm{kg}$ car has tyres with static $\mu = 0.52$ when gripping the road, but dynamic $\mu = 0.35$ in a skid. On a horizontal road, calculate the following.

Part A

Skidding

The maximum braking force which will not cause a skid.

Part B

Deceleration

The decelerating force if the driver brakes too hard and skids.

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Leaning Ladder

Subject & topics: Physics | Mechanics | Statics **Stage & difficulty:** A Level C2

*This problem involves **friction** , which is not covered in some Physics A Levels. For more information please check with your teacher.*

A uniform ladder of mass m and length l leans against a frictionless wall at an angle θ to the horizontal, with its bottom resting on a rough surface with a coefficient of friction μ .

If $\theta = 30\,^{\circ}$, what is the minimum value of μ such that the ladder does not slip?

- $\mu = \frac{1}{2\sqrt{3}}$
- The ladder will not slip, no matter what the value of μ is.
- $\mu = \frac{\sqrt{3}}{2}$
- $\mu = 1$
- $\mu = \frac{1}{2}$

Created for isaacphysics.org by Michael Conterio.

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Toppling Block

Subject & topics: Physics | Mechanics | Statics **Stage & difficulty:** A Level C2

*This problem involves **friction**, which is not covered in some Physics A Levels. For more information please check with your teacher.*

A rectangular block with a square base of side $10\,\mathrm{cm}$ rests on a rough horizontal surface. A slowly increasing horizontal force is applied to one vertical face. If this force is applied near the bottom of the face then as the force increases the block will slide before it topples. If the force is applied near the top then the block topples over before it starts to slide. When the force is applied $20\,\mathrm{cm}$ from the bottom, the block sometimes slides and sometimes topples.

Find the coefficient of friction between the block and the surface (to 2 significant figures).

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Three Cylinders

Subject & topics: Physics | Mechanics | Statics **Stage & difficulty:** A Level C3, University C1

*This problem involves **friction** , which is not covered in some Physics A Levels. For more information please check with your teacher.*

Two rough cylinders, each of mass m and radius r, are placed parallel to each other on a rough surface with their curved sides touching but not joined together. A third equivalent cylinder is balanced on top of these two, with its axis running in the same direction.

For this setup to be stable, what is the minimum coefficient of friction between the horizontal surface and the cylinders? Give your answer to three significant figures.

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